Transactions and Concurrency Control - lecture notes.

A transaction is a logical unit of work that must satisfy atomicity, consistency, isolation and durability.

Strict two-phase locking holds every exclusive lock until the transaction commits, which prevents cascading aborts.

A precedence graph has a node per transaction and an edge for each conflicting operation; a cycle means the schedule is not conflict serializable.

Deadlocks are detected with a wait-for graph and resolved by aborting a victim transaction.

Write-ahead logging forces log records to stable storage before the corresponding data pages, enabling undo and redo during recovery.

Snapshot isolation avoids read locks but permits write skew, an anomaly that serializable isolation prevents.